

Corporate Participants

- * Donald R. Young Aspen Aerogels, Inc. - President, CEO & Director
- * Laura J. Guerrant-Oiye Aspen Aerogels, Inc. - Vice President of IR & Corporate Communications
- * Ricardo C. Rodriguez Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Conference Call Participants

- * Alexander Eugene Potter
Piper Sandler & Co., Research Division - MD & Senior Research Analyst
- * Amit Dayal
H.C. Wainwright & Co, LLC, Research Division - MD of Equity Research & Senior Technology Analyst
- * Chip Moore
EF Hutton, Research Division - Managing Director
- * Christopher Curran Souther
B. Riley Securities, Inc., Research Division - Research Analyst
- * Colin William Rusch
Oppenheimer & Co. Inc., Research Division - MD & Senior Analyst
- * Eric Andrew Stine
Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst
- * Thomas Patrick Curran
Seaport Research Partners - Senior Analyst

Presentation

Operator

Good morning. Thank you for attending the Aspen Aerogels, Inc. Q2 2022 Financial Results Call. All lines will be muted during the presentation portion of the call with an opportunity for questions and answers at the end. I would now like to turn the conference over to your host, Laura Guerrant. Aspen's Vice President, Investor Relations and Corporate Communications. Thank you.

You may proceed, Ms. Guerrant.

Laura J. Guerrant-Oiye, Aspen Aerogels, Inc. - Vice President of Investor Relations & Corporate Communications

Thank you, Param. Good morning, and thank you for joining us for the Aspen Aerogels Fiscal Year 2022 second quarter financial results conference call. With us today are Don Young, President and CEO; and Ricardo Rodriguez, Chief Financial Officer. There are a few housekeeping items that I would like to address before turning the call over to Don. The press release announcing Aspen's financial results and business developments as well as a reconciliation of the management of non-GAAP financial measures compared to the most applicable U.S. generally accepted accounting principles or GAAP measures is available on the Investors Section of Aspen's website, www.aerogel.com.

Included in the press release is a summary statement of operations, a summary of balance sheet and a summary of key financial and operating statistics for the 2022 second quarter ended June 30, 2022. In addition, I'd like to highlight that we have uploaded to our website a slide deck that will accompany our conversation today. You can find the deck at the Investors section of our website.

An archive of today's webcast will be on our website for approximately 1 year. Please note that our discussion today will include forward-looking statements, including any statement regarding outlook, expectations, beliefs, projections, estimates, targets, prospects, business plans and any other statement that is not a historical fact. These forward-looking statements are subject to risks and uncertainties.

Aspen Aerogel's actual results may differ materially from those expressed in these forward-looking statements. A list of factors that could affect the company's actual results can be found in Aspen's press release issued yesterday, Page 1 of the presentation, and are discussed in more detail on the reports Aspen files with the SEC, particularly in the company's most recent annual report on Form 10-Q.

The company's press release issued yesterday and filings with the SEC can also be found in the Investors Section of Aspen's website. Forward-looking statements made today represent the company's views as of today, July 28, 2022, and Aspen Aerogels disclaims any obligation to update these forward-looking statements to reflect future events or circumstances.

During this call, we will refer to non-GAAP financial measures, including adjusted EBITDA. These financial measures are not prepared in accordance with GAAP. These non-GAAP financial measures are not intended to be considered in isolation or as a substitute for results prepared in accordance with GAAP.

The definitions and reconciliations of these non-GAAP financial measures to the most directly comparable GAAP financial measures and a discussion of why we present these non-GAAP financial measures are included in yesterday's press release.

And one final note, during the Q&A session in the interest of time, we ask that you please limit your questions to two questions at a time. If you have additional questions beyond the initial two, please get back into the queue, and we will get to all questions.

I'll now turn the call over to Don. Don?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Thanks, Laura. Good morning, everyone. Thank you for joining us for our Q2 2022 earnings call. I will kick things off with a progress report on our financing activities and recent business developments, and Ricardo will discuss business results and outlook.

We will conclude with a Q&A session. Over the past 16 months, we have raised over \$300 million to scale a company that is targeting revenue of \$1.6 billion and EBITDA of \$400 million as we seek to fully utilize our second aerogel manufacturing plant in the years to come. We have existing manufacturing capacity to meet our 2x revenue target of \$240 million for 2023 and expect to utilize Phase 1 of Plant 2 in order to meet our 3x revenue target of \$720 million for 2025.

We expect to continue to raise capital in order to build out the required infrastructure to capture our full opportunity. On June 29, we announced our decision not to proceed with concurrent public offerings of

common stock and convertible notes. While we had significant demand from investors, including a \$100 million indication of interest from Koch Strategic Platforms, overall market conditions were unfavorable.

And as such, the terms of the offerings were determined not to be in the best interest of Aspen shareholders. As we have said in the past and as is especially true in the current financial markets, we are taking in all of the above approach to financing our growth plan. As we explore a variety of prospective sources of capital, we have continued to focus on strategic investors who know our company and the markets we serve and who have the potential to make equity investments in the business, as Koch Strategic Platforms has done in the past and indicated its interest in doing so again in our June financing effort.

In addition to potential strategic investors and the public equity and debt markets we are engaged with government programs as potential 2023 sources of capital for Plant 2. We have applied for a \$150 million U.S. Department of Energy grant for advanced battery materials as part of the Bipartisan Infrastructure Act, which is designed to support U.S.-based companies dedicated to the electrification economy. We are also exploring other DOE programs that are focused on battery performance and safety.

The programs target American Manufacturing in an effort to address the resiliency of supply chains in the U.S. especially for projects in critical areas of sustainability, such as energy storage and related materials. While such DOE programs can take time and are unpredictable, we believe we are a very good candidate and that our pursuit is consistent with our all of the above approach to raising the necessary capital for us to execute our long-term strategy.

We also believe that as additional public market investors learn the Aspen story with its secular growth trends and sustainability-focused themes these investors and, of course, our existing investors will again be a source of capital for a fast-growing dynamic company like Aspen Aerogels.

Despite the challenging markets, we are confident that we will raise the necessary funds to execute on our strategy to become cash flow positive in 2024 and to create value for our shareholders. We appreciate your continued support.

Our business performance in Q2 remained strong despite lingering supply chain and staffing challenges. We had a strong start to the year in Q1 with 37% year-over-year revenue growth and Q2 built on that momentum with 44% year-over-year revenue growth. The first half performance reflected continued penetration in the EV market and high levels of activity in the energy industrial market.

With this commercial momentum and of course, our existing awarded programs with General Motors and Toyota, we are confident that we will have high year-over-year growth rates in 2022, and we believe that we are in a strong position to meet our \$240 million revenue target for 2023.

PyroThin thermal barrier revenue was \$11 million, was nearly \$11 million in Q2 compared to \$7.6 million in Q1 and less than \$7 million for all of 2021. We anticipate that multiyear thermal barrier revenue will be substantial, and we are investing now to meet the demand.

We also recognize that automotive OEMs will be impacted from time to time by their own supply chain challenges that could impact their growth ramps in any given period. We are trying to build in optionality to manage our overall revenue growth during this early stage of the EV megatrend by continuing to have a deep order book in the energy industrial side of our business.

The strong outlook for energy industrial is fueled by pent-up maintenance demand, high energy prices and strong long-term activity levels in LNG. This flexibility is a good example of the benefit of our strategy to leverage the Aerogel technology platform into a diverse set of large and dynamic markets.

I would also like to provide a brief update on Plant 2, our aerogel manufacturing facility under construction in Georgia, with long lead time items purchased starting in Q4 of last year, the first phase of Plant 2 remains on a projected time line for completion in late 2023. While the project is not immune to challenges related to supply chain disruptions and inflation, our team supported by Koch Project Solutions is focused on building a first-class aerogel manufacturing facility.

The first phase of Plant 2 targets \$650 million of annual revenue capacity and will bring our total annual revenue capacity to approximately \$900 million. As we approach full capacity utilization at this revenue level, we are targeting EBITDA of approximately \$225 million per year. With reports indicating that the auto industry is on track to invest \$0.5 trillion in the next 5 years, to make the transition to electric vehicles, we believe that we are well positioned to play an important role in battery performance and safety.

In addition to investments in Plant 2 in Georgia and in our high-volume assembly facility in Mexico, we continue to make important investments in people, systems and automation. These actions enable us to scale rapidly to transition to positive cash flow in 2024 and to monetize our investments in building our dynamic business.

We also continue to invest in our -- in the strategy to leverage our Aerogel technology platform into other high-value markets with sustainability themes. Our teams have been extremely productive this year, expanding our intellectual capital position across the Aerogel technology platform with over 60 new inventions captured in 33 patent families filed or in draft year-to-date.

We believe these investments in technology and new business development will continue to validate the richness of our aerogel technology platform and create significant shareholder value. We expect our carbon aerogel initiative within Aspen battery materials to be next in line to commercialize products. ABM has completed the development phase of its next-generation silicon added material that targets both lower costs and increased energy density and is now preparing qualifying materials for select automotive and battery OEMs.

While these enterprise-wide investments require capital and burden margins in the short term, we believe they set us up structurally so that we -- so that with full utilization of plant 2, we are positioned to generate \$1.6 billion of revenue and \$400 million of EBITDA and , at the same time, have the opportunity from new businesses for potential additional breakout value. We have important work still ahead of us, but I am encouraged by the progress we are making towards achieving our goals.

And finally, I would like to continue the practice of highlighting our ESG work during quarterly earnings calls. For the past 2 decades, ESG has been linked to the success of our business. It is a natural fit for us to explore new uses for our aerogel technology platform with the goal of improving the environmental performance and safety of our customers' products and processes. It is also at the core of our culture to respect and celebrate our employees by striving to create a diverse and inclusive environment.

We believe we have the responsibility to make a positive impact on our communities, and we are committed to creating a corporate culture that pursues its mission with the highest standards of integrity. We will be releasing Aspen's inaugural ESG highlights report and launching our ESG web page in the coming days, which will provide a comprehensive overview of our overall ESG strategy. We look forward to your feedback.

I will now turn the call over to Ricardo.

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Thank you, Don. I'll start on Slide 4 and our financial highlights for the second quarter. Starting with revenues, high demand across all our markets and the successful execution of various initiatives to increase our throughput have enabled us to deliver \$45.6 million of revenues in Q2. This is close to our company's quarterly revenue record of \$46.5 million in Q4 of 2019 and translates into 19% growth over the previous quarter and 44% growth year-over-year.

Year-to-date, the team has managed to grow revenue by 40% year-over-year to \$84 million. While we have mentioned that the EV thermal barrier opportunities materializing faster than originally expected, I will start by highlighting that our energy industrial revenues increased by 13% over the prior quarter to \$34.9 million. This is driven by very strong demand from our distributors as we continue to fulfill a meaningful backlog of delayed maintenance demand while continuing to displace competitors materials and new projects that are being built to higher efficiency standards with tighter time lines.

Delivering this level of growth in our core business while also increasing our EV thermal barrier revenues to \$10.8 million, a 42% quarterly increase demonstrates our team's continued ability to deliver the growth that is required to meet our long-term objectives.

Now I'll highlight our main expenses while reminding everyone that despite the growth that we've been able to deliver during the quarter, both of our business segments still aren't running at the annual revenue run rate that is needed to properly absorb our fixed expenses and meet our target profit objectives.

Accordingly, we continue to manage every cost element to ensure that all our recent productivity investments start delivering results. I'll return to this point in more detail and cover how implementing operational elements beyond simply having a higher revenue base will drive meaningful gross profit increases. These elements fortunately require no radical new process invention or development.

It is simply a matter of putting them in place and validating them. Material expenses of \$26 million for the quarter made up 57 percentage points of sales, which is over 10 percentage points higher than where we want these to be long term. This variance is currently driven primarily by the scrap levels required to accelerate our EV thermal barrier throughput with processes designed for lower volumes and nonrecurring inbound freight expenses to ensure an interrupted production.

Various product design changes aimed at simplifying our assembly and reducing the number of parts in the bill of materials of our EV thermal barriers will help us address these nonrecurring costs. Conversion costs of \$20.8 million reflected the last quarter of our thermal barrier assembly facility in Rhode Island, delivering most of our production. To illustrate the burden of this, our temporary hourly labor cost at this facility were \$4.5 million to deliver \$10.7 million of revenues during the quarter.

More broadly, the higher revenue run rate of our energy industrial business enabled our total conversion cost to decrease by 4 percentage points of sales quarter-over-quarter to 46% or \$20.8 million. Operating expenses which are key to delivering our revenue and profitability goals of 2023 and beyond were of \$21.4 million. This increased by \$4.6 million quarter-over-quarter versus an increase of \$6.7 million in Q1 over the prior quarter.

As we go into the second half of the year, our OpEx increases will be more modest and focused precisely on delivering 3 things: One, tangible productivity benefits through new process development and the implementation of systems that streamline our methods and drive productivity; two, new business awards through our EV thermal barrier technical sales efforts; and three, clear milestones in our R&D efforts. These include our silicon anode, carbon aerogel development efforts along with R&D efforts in our silica aerogel-based installation formulations.

These are the developments that drive lower chemical waste expenses through reformulation and enable throughput improvements such as longer rollings and faster line speeds while ensuring the same quality standards. Accordingly, our net loss increased to \$24 million or \$0.68 per share versus a net loss of \$6.7 million or \$0.23 per share in the same quarter of 2021.

Adjusted EBITDA was negative \$18.3 million in Q2 compared to negative \$3.4 million in Q2 of last year. As a reminder, we define adjusted EBITDA as net income or loss before interest, taxes, depreciation, amortization, stock-based compensation expenses, and other items that we do not believe are indicative of our core operating performance. In Q2, these other items included \$2.3 million of stock-based compensation and \$1.4 million of interest expense.

Next, I'll turn to cash flow and our balance sheet. Cash used in operations of \$10 million reflected our adjusted EBITDA of negative \$18.3 million and the reduction in operational cash needs of \$8.3 million that was mainly driven by a \$12.9 million increase in accounts payable. Capital expenditures during the quarter of \$37.9 million included the ground clearing foundation, and building of Plant 2, assembly equipment for a higher-volume thermal barrier operations and the R&D lab upgrades for our carbon aerogel battery material efforts.

As progress remains on track for Plant 2 to come online at the end of 2023, we have capitalized \$62.5 million through the end of Q2 towards it. Cash provided by financing activities of \$4.9 million during Q2 included \$4.8 million of net proceeds from our ATM offering transactions at a gross average price of \$20.18

per share. Year-to-date, we have raised \$67.9 million opportunistically through ATM offering transactions at an average price of \$14.46 per share.

As Don mentioned earlier, we're employing in all of the above approach to raising capital. And as we consider all markets and avenues, the ATM offerings play a very small role in our broader financing efforts. We ended the quarter with \$162.2 million of cash, no borrowings under our revolving credit facility and shareholders' equity of \$164.7 million.

Our outlook for the year remains unchanged. We are geared to deliver revenues of \$180 million with potential upside to \$200 million, and net loss in the range of \$79.8 million and \$86.8 million and adjusted EBITDA in the range of negative \$55 million and \$62 million.

Our capital expenditures for the year are expected to range between \$250 million and \$300 million, delivering \$100 million to \$120 million of revenues during the second half of the year implies a 10% to 32% increase in our Q2 revenue run rate. This increase is subject to various external factors beyond our control, such as our thermal barrier customers' ability to fulfill their stated vehicle production volumes and the supply chain of our raw materials, such as silanes, batting, CO2 and the local labor market, particularly for our Aerogel facility in Rhode Island.

We are proactively managing our supply chain risks and have ensured that we're supplied of silanes and batting to execute our production plans. Recent nationwide CO2 shortages in a tight labor market, post the highest near-term risks to our continued throughput increased plants.

In managing CapEx we aren't seeing the same inflationary pressures of Q1 and are cautiously ensuring that we pay the latest prices for some of the key commodities going into Plant 2 versus the cost at which some contractors built inventory earlier in the year.

Turning over to Slide 5 and going back to my earlier point on profitability. During the quarter, our EV thermal barriers segment run at a roughly \$40 million annual run rate. While our energy industrial business ran at a roughly \$140 million annual revenue run rate. At these run rates, we're still not quite getting the most out of our overhead and fixed asset base.

Above our run rate of \$150 million per year, our energy industrial business can deliver 20% plus gross margins. Our EV thermal barriers segment requires an annual revenue run rate of around \$120 million to deliver 15% plus gross margins as we ramp up production in Mexico.

The introduction of simpler designs and more automation will help us lower this run rate requirement. Our EV thermal barrier business has been quoted to deliver our target margins at capacity, and our energy industrial pricing environment is favorable enough to support price increases that will take effect on every order fulfilled in 2023.

Beyond fixed cost absorption, our operating plan includes several initiatives that we're executing on both segments to pay our path to 35% gross margins at full capacity. On this slide, we'd like to illustrate for you how the implementation of specific initiatives in our operating plan will drive gross margin improvements as our revenue growth at the same time.

As I mentioned earlier, we're not reinventing processes and don't carry technical risk as we do this. It's simply a matter of getting these initiatives done as we have year-to-date with the successful start-up of our thermal barrier facility for assembly in Mexico. Our EV thermal barrier business is undergoing a meaningful operational transition.

As we mentioned earlier, our production over the past 3 quarters has been delivered with a set of low-volume processes in a manual facility in Rhode Island. To illustrate this, the number of thermal barrier assembly hourly staff in this facility increased from 278 at the end of November of 2021 to a peak of 428 in June of this year.

Today, we are back down to 280 people and have approximately 600 people in Mexico, ready for the volumes of the rest of this year and 2023. As we move this work to Mexico and introduce processes designed

for higher volumes, our conversion costs are going to decrease significantly and improve our gross margins by over 30 percentage points in 2023.

For Q1 of next year, we also have planned various automation processes for the encapsulation of the Aerogel and the assembly of the thermal barrier parts. These will further reduce our conversion costs by reducing cycle times, the labor cost per part and at the same time, increase our production yields. The reduction in part complexity by implementing various engineering changes, working with our customers is also streamlining our processes, aiding with pricing and driving productivity over the next 2 years.

In our energy industrial business, portfolio management, longer product runs with less changeovers and reformulations in our base chemistry to reduce our chemical waste expenses will continue to drive conversion cost reductions. Longer insulation production rolling and faster line speeds will increase productivity by Q4 of this year.

Our recently expanded supply chain team is also more than making up for the incremental fixed costs by streamlining our logistics and driving the negotiation of raw materials to our larger scale.

Plant 2 in Georgia will also improve our cost position overall with a linear process that will require little manual material handling and circular processes for key chemicals such as CO₂ and ethanol. This will reduce our material costs significantly, along with our environmental impact and the cost of this impact. Our target margins also factor in the DNA of Plant 2 that will kick in as it comes online in 2024, and we grow into its full capacity.

As 1 can see here, our plans to not just rely on higher revenues to improve profitability. Most of the improvements come from the successful implementation of productivity improvements and the launch of operations in flexible and more productive locations with processes designed for higher volumes. This plan is well within the capabilities of our current team and recent milestones keep us motivated to continue on our path to 35% gross margins of capacity.

Turning over to Slide 6. We would like to provide a brief market update on what we're seeing in the automotive EV market, how these dynamics validate our strategy and the effect that these are having on our commercial discussions with the world's most relevant OEMs.

Here, you can see that 7 of the top 8 OEMs that are expected to command over 65% of the EV market are focused on high energy density, high nickel content chemistries. They are also focused on prismatic or pouch cell form factors. These configurations are most compatible with cell-to-cell barriers like PyroThin. Several OEMs are also evaluating lower energy density chemistries such as LFP, and we are actively quoting LFP programs as well.

When we look at the market, we see PyroThin incompatible with at least 19 million vehicles. And what Piper Sandler forecasts to be a 24.7 million unit market in 2025. We are creating the market for aerogel-based cell-to-cell barriers. We do this by leveraging the innovative thermal isolation, fireproofing and mechanical properties of our material, having a system-level approach and building unrivaled scale.

All in all, the size of the market and PyroThin positioning is a unique solution presents an increasingly compelling opportunity.

Turning over to Slide 7. Here, we can see some other encouraging news and a summary of some of the most relevant UN global transportation requirement updates for EV safety that are bringing additional focus to the issue of thermal propagation and thermal runaway around the world.

This issue is currently not at the top of consumers' minds when EVs make up less than 10% of the new vehicle market, but will become more relevant as EVs make up a higher share of the global vehicle part and more than 25% of global new vehicle sales in 2025. The update to the global transportation requirements is focused on providing drivers with a 5-minute warning intended to give them time to evacuate the vehicle in the presence of a danger caused by thermal propagation or runaway in a single cell.

Current battery management systems can already trigger this warning, but the hard part is enabling 5 minutes or more of safety. With PyroThin, we can enable OEMs to turn what is currently a catastrophic event into a

serviceable event. The GTR updates became mandatory new vehicle regulation for new vehicles last year in China, South Korea and Japan.

They are being considered for implementation in the U.S. this year and come into effect in India next year, helping OEMs to not just meet these warning requirements but provide real protection in the rare case of thermal propagation and runaway and proactively developing the capabilities and scale to meet their standards is ensuring that we maintain our lead position as we create this market.

In Slide 8, the results of our market building efforts are illustrated where we're attempting to provide you with an accurate and constructive view of our commercial efforts. In this chart, the size of the circle is the vehicle volume in millions that Piper Sandler is forecasting for these OEMs in 2025, and their placement on the map is their approximate headquarters location.

The color of the circle then determines whether we have been awarded business by that OEM are actively quoting business, undergoing testing or not active with that OEM. It's no surprise that GM and Toyota are the red circles with 2.3 million vehicles in North America and 0.5 million vehicles in Japan, respectively. You can see that our team has successfully entered the quoting stages over the last 18 months with the largest customers in Europe and Asia that will be relevant on a global basis in 2025.

We feel confident in their ability to convert this activity into additional awards over the next 12 months and are glad to see the value of these quotes surpass the value of our current awards with GM and Toyota. As you all know, the timing of these awards is linked to an OEM's readiness to award the business and will be as forthcoming as possible with our communication as these awards materialize.

Assessing the value and probability of these quotes takes a path for us to become a company with \$1.6 billion of annual revenue capacity, delivering over \$400 million of EBITDA shortly after we achieve our near-term objectives through 2025.

With that, I'm happy to turn the call back to Don.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Thank you, Ricardo. Before we turn to the Q&A session, I would like to reiterate 2 important points. First, both our EV and energy industrial businesses are going strong and are providing a favorable backdrop for us to reach our 2023 and 2025 revenue targets.

And second, we are laser-focused on turning the investments that we are making today into a business that is cash flow positive in 2024 and is on the way to its longer-term goal of \$1.6 billion of revenue and \$400 million of EBITDA. Param, Let's turn to the Q&A.

Question And Answer

Operator

Absolutely. [Operator Instructions]. Our first question comes from the line of Eric Stine with Craig Hallum. Eric.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst

Good morning, everyone. So first, thanks on Slide 8. I mean that's a great slide, very informative. So thanks for including that. Maybe just thinking about the acceleration in activity on the EV side, I'm just curious how scarcity value is playing into that and how it's playing into discussions and also plans that you're making on your side as you work towards Phase 1 here of Plant 2?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Well, let me just start by saying that it is important that we demonstrate our ability to build out our capacity. As we've said many times, we have the capacity in place to meet our 2023 targets, and we'll be relying on

Phase 1 for our 2025 revenue targets. But it's even more than that. We know that even in the 2025 time frame, if you take energy, industrial, General Motors and Toyota, it represents a significant majority of that capacity.

So it's important that we continue to demonstrate to this next wave of OEMs that will have the capability of supplying them. And we think we're doing a good job of communicating that.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst

So today, I mean, it sounds like today, that's not necessarily a concern of OEMs, although I would think it is something that gets them to -- given the importance of this issue to want to maybe be further up in line in terms of the decisions they make around the quoting that you've done?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

I think that's a fair -- I think that's fair, Eric. I think that's a fair sort of instinct that they have. And it's --again, it's part of our communications as we go out and win this next set of awards.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst

Yes. Okay. And then just on the second one, just the fire containment that is -- it seems to be a pretty important thing that you're able to do? I mean, what type of learners in the industry are you seeing -- I mean how does that play into the acceleration of activity as well, making it more of a service event rather than catastrophic as you put it?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Well, the way I think about that, and Ricardo, I think, articulated well, both in the script and on the related slide that we're seeing some of the regulatory standards begin to be implemented. Those are -- I think those are positive for us, for sure. Most of the OEMs we're working with are setting a higher standard for safety than those baseline requirements. And we've discussed openly in the past that our work and the design of our products are striving not only to slow the propagation, but to isolate the bad cell.

And that is a very dramatic mindset change from having that catastrophic fire that is well understood to again, to a service event that impacts a cell or a module itself that can be replaced, again, in more of a service-oriented manner than the replacement of the automobile overall.

So we think those standards are excellent. We've never really relied on regulatory aspects to drive our business, but we think those are foundational and very positive for our business.

Operator

Our next question comes from the line of Alex Potter with Piper Sandler.

Alexander Eugene Potter, Piper Sandler & Co., Research Division - MD & Senior Research Analyst

Great. So I just had a question. This is an interesting slide. Slide 8, it's encouraging to see all that quoting activity, including on, I guess, land masses other than North America. But one of the questions I had on that is obviously, Altium is going to be primarily a North American platform. But some of these other say, European or Asian OEMs, clearly, they have their own platforms, and they have their own plans in those regions.

A lot of those places are pretty far from Georgia. So how much of a concern is that for some of these customers as people are looking more and more localizing battery supply chains? Are you able to supply all of their needs out of Georgia? Are any of them insisting on having more localized capacity either in Europe or Asia to support their own ramps?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes, it's a very good observation. The way we're thinking about it today and the way we've communicated it to the OEMs is that we will -- we will have our first \$1.6 billion of revenue capacity here in North America between our Rhode Island and Georgia plants.

But what we are also committing to is to have fabrication assembly facilities regionally. And just as we have our North American operation, principally expanding very rapidly in Mexico. It's clear to us that a European OEM will want to have that capability and move that supply chain -- that aspect of the supply chain closer to their own assembly plants.

So -- and I guess I would also just say that while we locate the circle, if you will, in their home base, -- many of these companies are global in their nature, so in their locations. So the regional locations, I guess, I should say. And so anyway, so we are mindful of that, Alex.

And I think depending on how our demand builds out over time and as we fully utilize Plant 2, we will certainly be mindful of that mix, geographic mix of our business as we think about building a second plant. Today, we are highly focused on Phase 1 and on Phase 2 of Plant 2 and getting to that \$1.6 billion of revenue and \$400 million of EBITDA target.

Alexander Eugene Potter, Piper Sandler & Co., Research Division - MD & Senior Research Analyst

Great. Perfect. Yes, one thing at a time. Okay. So the second question, you alluded to this a couple of times in the prepared remarks. I'm hoping maybe you can elaborate on a little bit. It sounds like you may be experiencing some staffing bottlenecks.

It sounded like that was primarily in Rhode Island, but it also sounded like you had -- now that the Mexico facility has opened up, you're able to, I guess, downsize a bit in Rhode Island. So if you could just maybe elaborate on hiring, staffing, any potential problems you're having either in Rhode Island or in Georgia, that would be helpful as well.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes. Thank you, Alex. We have improved the staffing of our aerogel manufacturing facility here over the course of the past 6 months. It was a bit of a struggle for us the latter part of last year, Q4 in particular, coming into this year and Q2. And here in July, we have made good progress in staffing that facility.

And so it is less of an issue for us. The -- we do -- and Ricardo alluded to this really in the margin walk and what are we doing to improve. We have redundant activities going on today as we ramp. And I think the Rhode Island assembly to the Mexico assembly facilities, Mexico is really meant for the high volumes that we're ramping up to here.

Right now, we have sort of both of them operating, that won't be required going forward. I will always have some capability here, but it will be more in the prototyping or very early-stage work that we do.

So being able to eliminate those redundant activities, being able to eliminate -- to get to some of the nonrecurring activities, get those off of our income statement. We'll improve our gross margins improve significantly. And we're very focused on that.

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Yes. If I may add another key thing here is, particularly when it comes to the aerogel plant in Rhode Island, our HR team and the operations team have come up with an incentive scheme aimed at managing retention. And we -- I know I mentioned that we currently have over 200 people still on the thermal barrier assembly side in Rhode Island that will drop here to less than 100 before the end of this month as well.

So that transition is actually a pretty quick one in Q3. And so while we're highlighting the risk, we do think that the team has implemented enough measures to get it under control.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes.

Operator

Our next question comes from the line of Chris Souther with B. Riley.

Christopher Curran Souther, B. Riley Securities, Inc., Research Division - Research Analyst

Could you provide a bit more color on the commercial progress slide? Just how should we think about quoting and testing versus the way you used to discuss the 3 stages? Should we think about testing or quoting as kind of the last step before Toyota and GM orders came in and maybe just the time lines that it took from the different steps with GM and then Toyota and how that kind of time line either shrunk or was consistent would be helpful.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Well, certainly, the quoting phase is very much in the Stage 3 part of our earlier kind of Stage 1, Stage 2, Stage 3 serve concept. And I think testing is solidly in probably the later stages of 2 as we, again, use those concepts. But when we are quoting, we have gone through a significant amount of testing and qualification and around quality systems right through to product performance and specifications.

So this is pretty late stage in our development work with a given OEM. So where you see the blue circles on that Slide 8 that is definitely from Stage 3 work that we're trying to translate into awards here over the course of the coming 2, 3, 4 quarters.

Christopher Curran Souther, B. Riley Securities, Inc., Research Division - Research Analyst

Got it. Okay. And then just kind of looking at the awarded OEM forecast, you're expecting here and the ranges you've talked about are content per vehicle. It looks like you guys are being really conservative with that \$540 million 2025 PyroThin target on either the volumes or the content per vehicle.

So maybe you could just kind of walk through where you guys think you are setting yourself up particularly nicely because obviously, if it's towards the midpoint of that, we're already looking at filling up Phase 1 of Plant 2 pretty quickly there.

So I just wanted to get a sense of how we should think about that. And then our content per vehicle still kind of coming in that range you guys had initially talked about or not initially, but subsequently, we're talking about.

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Yes. I mean this is an interesting one. So we are seeing a difference in the content per vehicle between prismatic cell configurations and those OEMs where our technology is going in between pouch cells. And so, for example, Altium is running pouch cells. And so their content per vehicle is some are well above \$1,000 a vehicle, but we think it will settle to around \$600 a vehicle.

And you're right. I mean when we take all the external assessments of GM's volume or Piper Sandler's here on Slide 8 at [2.3 million] in 2025. That would put us well above the \$540 million that we've communicated for thermal barriers so far.

We are going through making our own assessment of where GM's volumes will truly lie and where the CPV will land. Right now, we still think that the tripling of revenue with that mix of thermal barrier and energy industrial revenues in 2025 is still accurate to the best of our knowledge and with all the purview of the information that we have.

But yes, I mean, there's definitely a chance to -- in which we could very well find ourselves in a situation here, having to accelerate the buildup of Phase 2 in order to accommodate an expansion of the current awards and these other awards that the commercial team is on the final stages of.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

I think one other thing, Chris, that I would just mention is that if you look across the spectrum of General Motors and the vehicle, the model that we're on with Toyota. General Motors has started off with some fairly large vehicles, frankly. And if you look at their rollout here that they've articulated very clearly over the course of 2023, 2024, 2025, you get quite a broad range of vehicles and including some smaller, higher-volume vehicles.

And we know that those vehicles have fewer modules making up the battery pack and CPV sort of translate from there. So your question is a very good one, and we're mindful of it. We've, I think, continued to try to have people focus on CPV in that \$300 and \$350 kind of range, and even though the numbers are higher than that today.

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

And I guess this also goes back to why Phase 1 was scoped out the way it is because when we look at the CapEx of Phase 1, it really is around \$200 million of equipment inside of a \$375 million building that is already equipped to handle both phases. And so then we're able to bring in Phase 2 within around 12 months' notice at any given point in time.

And so here, as these quotes convert into awards, we could very well accelerate Phase 2. And that's why our mind really is at this \$1.6 billion of revenue level and \$400 million of EBITDA as we look into the future.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes. That spending in Phase 1 to create the infrastructure to support both phases Phase 2. We estimate to cost an additional \$125 million. But clearly, we're spending in advance to be faster in bringing up that second wave of capacity in Plant 2. And I would say that's another example, and Ricardo and I talked about it in our scripts of -- the investments that we're making now to be able to handle the volumes that we see in 2023, '4 and '5.

And whether those are in people and processes and automation, we are making those investments to a great extent here in 2022 to prepare ourselves for those higher volumes. And we understand that it's pinching our margins today, but we do believe in combination with the actions that Ricardo articulated in his script, doing those 2 things, good business practices and in our margin walk and leveraging the greater volumes. The math works out very favorably for us as we strive for those 35% gross margins that we've articulated.

Christopher Curran Souther, B. Riley Securities, Inc., Research Division - Research Analyst

Okay. No, that's very helpful. And just if you kind of add up all the people that you're already quoting, 1.6 is probably not enough here. If you were to kind of win everything, right? So I'm curious how you would think about kind of capacity beyond that? Is there additional space around the facility where you could kind of have a Phase 3 there? Or is it something that you think you'd kind of look to build elsewhere to kind of piggyback on someone else's question.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes. That's -- these are good questions. And from that, we do try to focus on preparing for success, and we feel we're doing that with a lot of the investments that we're making here. I believe that when we have plant to fully utilize that \$1.6 billion revenue level and generating significant amount of cash in the business that we are likely to want to build our third aerogel manufacturing facility elsewhere and quite possibly in an international location, getting back a little bit to Alex's question as well that those OEMs may very well want to shorten that supply chain. And we think that will make a lot of sense for us to do it to think about it that in that manner.

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Yes. I mean another thing worth noting on Slide 8 is that the -- I mean these aren't exactly the volumes that we're quoting with these OEMs, right? These are the market sizing of how many EVs, the 5% forecast expects these OEMs to sell in 2025.

In many of these quotes, we were quoting a platform that has a subset of their total EV volumes or a nameplate that is being rolled out past 2025 as well. So we think we can manage within the 1.6 billion pretty well here over the next 3 to 4 years.

Operator [26]

Our next question comes from the line of Jason Burnoff on behalf of Colin Rusch with Oppenheimer.

Colin William Rusch, Oppenheimer & Co. Inc., Research Division - MD & Senior Analyst

Sorry, this is actually Colin. I'm on the line. So the energy industrial revenue here and the diversity of customers, it looks like you guys are making some good progress in terms of expanding that customer base and then also the visibility on growth from this year to next year, which looks pretty substantial.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

It is really rewarding -- we have -- these are -- this is our initial market. Obviously, we have an extremely strong team around the world serving that business. It continues to be a business while we have tremendous growth here in the U.S. It continues to be a majority of the business outside the U.S.

So I really like the footprint, if you will, of the business geographically. I would just say that the diversity is also quite outstanding in the array of applications that we're serving. Maintenance work continues to be an important part of that business and maintenance work has always felt to us a lot like baseload.

Apart from the COVID interruption, that business grew nearly every quarter since we introduced those products on the maintenance side. And then we get the swings a bit more on the project side. And we're seeing interesting project opportunities in our traditional sort of pipe and pipe subsea pipeline activities, but also, of course, in the LNG business, both in the liquefaction side and on the receiving terminal side in all regions, really.

And so we're really excited about that business. Sometimes, I think it gets lost in our discussions about the EV mega trend for sure, but it is a significant business that we think that we can continue to grow. And we also believe that our products focused on efficiency, asset resiliency and safety are spot on as these facilities. Think about their own ESG goals and sustainability commitments. So again, I appreciate the question. It's really an important part of our business.

Colin William Rusch, Oppenheimer & Co. Inc., Research Division - MD & Senior Analyst

Okay. Excellent. And then on the CapEx numbers, it looks like you spent about \$66 million, \$67 million through the end of 2Q. And you've got a fairly substantial amount left to go, 200 or a little bit more for the balance of your plan this year. I guess, can you talk a little bit about the cadence of how that money is going to flow out? And any of it that you can defer into next year without any real impact to time frames.

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Yes. I mean there's a good chance to defer some of the Q4 spend, which would be of around \$150 million. And then this quarter is looking like it's going to be anywhere between \$70 million to \$100 million of CapEx. We're actually seeing the expenses come in later than we originally planned. And yet, there is a good chance for around 15% to 20% of our total CapEx budget for this year to spill over into next year.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes, we're very focused on that, Colin. As you can imagine, we want to balance, obviously, our balance sheet and the importance of building Plant 2 in a timely way. So we're very focused on making sure we're in good shape on both of those fronts.

Operator

Our next question comes from the line of Tom Curran with Seaport Global Holdings.

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst

For thermal barriers, when I model guidance, it suggests you should hit that annual revenue run rate of \$120 million over the second half of next year, so potentially just 12 to 15 months out and you expect to be able to achieve a gross margin of 15% at that run rate.

Ricardo, would you please revisit and bridge for us the expected upswing in gross margin from the steep negative level being incurred at \$40 million to 15% and \$120 million. Could you please just break down and quantify each driver's expected contribution to that target margin improvement?

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Sure. So I mean, conversion, I mentioned in my script that over 30% of an improvement, at least in 2023, when we look at the full year, we think that will be closer to 35% as we think about that. Then our ability to improve the way we absorb fixed expenses, that's about another 10% improvement. Then some of the engineering changes that we're implementing are actually reducing the overall material cost as well.

And that's -- I think we're being a little bit conservative as we assess that, but that could be anywhere from 1 to 5 percentage points of sales here as that shakes out.

Logistics, as we go from shipping Aerogel from -- as we go for shipping less Aerogel because we're improving our yields in Mexico, that's about a 6 percentage point improvement in 2023.

And so when we put these all together that's what gets us to that positive 15 plus on a run rate basis closer to the third quarter of next year. I think that's how it would break down. I mean, I think the negative 66% of Q2 can't be totally walked to the 2023 point because of the variable labor expense -- of the temporary labor expenses that I outlined, right? So if we take that out...

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst

Right. And that's why it wasn't in that annualized \$40 million was at that full negative 66% just steeply negative, right?

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Exactly, so I mean, I think as we look at Q2, I mean, Q2 was better -- or Q1 was better, Q2 got worse. Q3 is probably going to be about the same, and then we'll see our improvements really kick in, in Q4.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

And it really goes to a lot of the...

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst

The first half of 2023 should really be the big inflection point? In terms of...

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Correct. Yes...

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst

The crossover into profitable and then double-digit gross margin.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes. We will have eliminated the majority of some of the redundant operations that we have. We will have addressed a lot of the nonrecurring expenses that we have as we scale up right now. And those are big drivers in the numbers as well, especially compared to where we are here in Q2 and Q3.

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst

Great. Helpful. And then for the \$150 million DOE grant application, what would be the timing and nature of the next milestone on the path to approval? And if you do qualify and it certainly seems to me that you should, you're certainly an ideal candidate. When should the grant be extended?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

It would have a 2023 impact on our capital. And this will -- the process is going to play out here over the course of the next 6 months, Tom. And these things are -- these things can be a little opaque, but we have good advisers and we're very engaged, and we think we have an excellent application, if you will, for the grant.

We're well engaged with the DOE. And so again, I think it's a little hard to -- I think I said in my script that these things play out in a little bit of an unpredictable manner. But again, I think we are a very good candidate, and it fits neatly into our all of the above strategy. And we're encouraged, I think, by where we stand in this. But we will give updates on -- certainly on a quarterly basis as we make progress through the process that the DOE has outlined.

Operator

Thank you for your question. [Operator Instructions] Our next question comes from the line of Amit Dayal with H.C. Wainwright.

Amit Dayal, H.C. Wainwright & Co, LLC, Research Division - MD of Equity Research & Senior Technology Analyst

Most of my questions already answered. Just on the battery development side. Any updates for us? Are you any closer to moving forward with testing, et cetera, with customers?

Ricardo C. Rodriguez, Aspen Aerogels, Inc. - Senior VP, CFO & Treasurer

Yes. I mean on that end, we've pretty much exhausted the testing that we can do in coin cells and are transitioning to testing on larger pouch cells. And as we do that, I mean, obviously, doing that with some partners whose core competency is making larger form factor cells is ideal, and that's what the team is right now focused on really trying to find the right relationship so that we can accelerate our testing just given the encouraging results that we've seen on coin cells. And as we do that, that's when the -- really the pace of validation will ramp up here in the second half as we try to meet our goals for the year.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

We've made in the past 12 months some meaningful investments in that team and equipment and that is paying off. We're -- as I said in my script, we are now able and in the process around producing what we refer to as qualifying materials, and those are destined for a select group of automotive and battery OEMs.

And so we'll see that play out here in the coming quarters. This is not a fast process. We don't underestimate how this works and the progress that we're making technically, we think, is outstanding, both from a performance and a cost point of view. And we also really believe in the market itself, and that is to say the importance of introducing silicon in greater percentages into lithium-ion batteries as a means of improving energy density and drive range.

Operator

Our next question comes from the line of Chip Moore with EF Hutton.

Chip Moore, EF Hutton, Research Division - Managing Director

I wanted to follow up on the DOE funding. I agree you should be a great candidate there for grant. I'm actually more curious about Altium securing their conditional loan. Do you think there's potential there to accelerate PyroThin sales? Or do you think that's largely contemplated in their existing plans?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Yes. That was terrific. That was part of a different program, but the same themes, right? And that was a loan and we've applied for a grant. But again, we were encouraged by what General Motors and LG, I think, together, we're awarded in that process because I think we haven't upped our numbers to GM because of that. But again, we just find it really encouraging that, that business is well supported, both internally by General Motors and LG and externally in such programs is \$2.5 billion from the DOE.

Chip Moore, EF Hutton, Research Division - Managing Director

Understood. That's helpful. Just one more for me. On Battery Materials. You talked about some qualifying materials. Can you give us maybe a sense of how much of a replacement for graphite sort of these first materials could be and maybe sort of a sense of path to future iterations.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Well, we're working through much of this today. Our team has articulated goals of 20% to 40% replacement - or I should say, silicon content. And that's a fairly large range, obviously. And a lot of times, that range has as much to do with the overall chemistry and system, if you will, as it does with the capabilities of our material.

So we think that range will vary according to the cell manufacturer and the battery design itself. But meaningful improvements relative to the relatively small amount of silicon in some of today's lithium-ion batteries, single-digit typically at best.

Operator

Thank you for your questions. There are currently no further questions waiting. So I will pass the conference back to Don Young for closing remarks. Thank you.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director

Thank you, Pam. Thanks for your help. We appreciate everyone's interest in Aspen Aerogels. We're very excited about the work we're doing and finishing the year strongly, and we look forward to reporting out our third quarter 2022 results in October. Happy well. Have a good day. Thanks so much.

Operator

This concludes today's conference call. Thank you for your participation. You may now disconnect your lines.